Agency and Learning Efficiency as the Structural Invariants of Intelligence

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Abstract

We prove that for any entity with structure (K, A, σ, γ) persisting under bounded active context, survival pressure, and positive novelty rate $\nu(E) > 0$: (1) agency $\gamma > 0$ is necessary: the lineage argument shows that $\gamma = 0$ lineages contract and are eliminated; (2) learning efficiency $\eta_M > 0$ is necessary: without it, the active context L_n fills without relief and inference degrades to failure; (3) the action space A (informed actions) and the higher-order function space M (learning) are type-irreducible: no learning process changes a function's type signature; (4) every other property of the entity can equal zero for some persistent entity; (5) therefore $I(E) = \gamma \cdot \eta_M$ is the unique measure of intelligence, derived not stipulated. This completes the structural theory: from bounded context alone, the complete architecture of intelligence follows as necessary consequences.

1 Introduction

This paper completes a chain of results deriving the architecture of intelligence from a single generative constraint: bounded active context.

Paper 1 [13] established that any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges. Paper 2 [14] established that bounded context under growing action count requires escalation to enable top-down allocation. Paper 3 [15] established that survival pressure makes knowledge and action inseparable, with an asymmetric retention criterion. Paper 4 [16] established that agency with feedback from the world produces gradient descent on uncertainty, with convergence, and that cost-of-failure generates an encoding hierarchy with permanence.

One question remains: **what properties must every persisting entity have?** We prove that the answer is exactly two: agency $\gamma > 0$ and learning efficiency $\eta_M > 0$. Their product is intelligence.

This result engages a long-standing difficulty. Dupoux, LeCun, and Malik [6] deliberately avoid defining intelligence, instead characterizing it as the capacity for autonomous learning. We derive the definition they avoided giving. Hutter [11] and Legg and Hutter [12] define intelligence as average performance over computable environments; we define it as rate of improvement, independent of current performance level. Chollet [4] measures skill-acquisition efficiency, structurally close to η_M but treated as a benchmark metric rather than a structural invariant. Schmidhuber [19] identifies curiosity as compression progress, which corresponds to η_M, but treats drive as a design objective rather than a necessary condition of persistence. Friston's free energy principle [9] derives minimization behavior from self-organization but does not establish the minimization *imperative*: why $\gamma > 0$ is structurally necessary.

The contribution of this paper is to prove, via a selection argument on lineages, that $\gamma > 0$ and $\eta_M > 0$ are the unique structural invariants of persistence, and that $I(E) = \gamma \cdot \eta_M$ is the only consistent intelligence measure.

2 Formal Framework

We build on the definitions of Papers 1–4 and introduce the structures needed for the present results. Throughout, W is a measurable space of world states, K is a knowledge base of propositions about W, C_n is the bounded active context capacity, A is the action space (actions informed by knowledge), and $U(w, K) = -\log P(w \mid K)$ is the uncertainty of K about state w [21, 5].

Definition 1 (Novelty Rate)**.** The *novelty rate* $\nu(E)$ of entity E in world W is the rate at which E encounters world states w where $U(w, K) > 0$:

$$\nu(E) = \lim_{T\to\infty} \frac{1}{T}\big|\{t \le T : U(w_t, K(t)) > 0\}\big|$$

Novelty rate is a property of the gap between W and K, not of W alone. A world with high entropy but an entity whose K covers all reachable states has $\nu = 0$. A world with low entropy but an entity with impoverished K has $\nu > 0$. The distinction matters: entropy is a property of the environment; novelty is a property of the entity–environment coupling.

Definition 2 (Free Context)**.** The *free context* of entity E at time t is the active context capacity not consumed by unresolved content:

$$C_{\text{free}}(t) = C_n - |\text{ctx}_{\text{unresolved}}(t)|$$

where $\text{ctx}_{\text{unresolved}}(t)$ is the set of propositions in active context that have not yet been encoded at a lower level or resolved by M. Free context is the entity's reserve capacity for handling novel states. Bounding uncertainty on known problems frees context; novel problems consume it. The architecture derived in Papers 1–4, graph structure, escalation, encoding permanence, exists to maximize C_{free}: each result describes a mechanism that moves resolved content out of L_n, preserving capacity for what the entity has not yet encountered.

Definition 3 (Higher-Order Function Space M)**.** Let $A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$ be the action space of informed actions, where W_{obs} denotes observable world states and $\Delta\mathcal{A}$ the simplex over available actions. The *higher-order function space* is

$$M : (A, K) \times L \;\to\; (A, K)$$

where L is a learning signal (feedback from W following action). M is categorically different from A: elements of A map knowledge and observations to action distributions; elements of M map action–knowledge pairs and learning signals to updated action–knowledge pairs.

M contains whatever adaptive capacity has not yet been committed to specific knowledge or action. As the entity learns, components of M resolve into concrete elements of K or A—the same normalization by which shared propositions precipitate from action into knowledge [15]. What remains in M is the entity's capacity to produce further such resolutions.

The hierarchy is open upward: $M^{(2)}$ operates on $M^{(1)}$, yielding meta-learning; $M^{(n+1)}$ operates on $M^{(n)}$. No natural termination point is established (see Section 8).

Definition 4 (M Efficiency)**.** Let $G(a, K) = I(p_a; W \mid K)$ denote the mutual information between the feedback from action a and the true world state W, conditional on knowledge K—the *grounding* of a. Given a learning event where $(A', K') = M((A, K), L)$, define the *learning efficiency*:

$$\eta_M \;=\; \mathbb{E}_{L|\text{engaged}}\left[\frac{\max_{a' \in A'} G(a', K') - \max_{a \in A} G(a, K)}{|L|}\right]$$

The expectation is taken over learning signals L *conditional on the entity engaging* M (i.e., conditional on the gate γ opening). This conditioning is essential: γ may be directed (the entity may selectively engage on high-quality signals), so $\mathbb{E}[\cdot \mid \text{engaged}]$ may differ from the unconditional $\mathbb{E}_L[\cdot]$. Conditioning on engagement ensures that the product $\gamma \cdot \eta_M$ is the expected improvement rate by the law of total expectation, regardless of whether γ is content-blind or directed.

Both numerator and denominator are measured in bits; η_M is dimensionless. The three regimes:

- $\eta_M > 0$: the entity learns: grounding improves per unit of learning signal, conditional on engagement.
- $\eta_M = 0$: engagement produces no improvement: M is engaged but yields no grounding gain.
- $\eta_M < 0$ temporarily: exploration, creative destruction, locally irrational steps that may increase long-run survival. Persistently $\eta_M < 0$: the entity does not persist.

Remark 1 (Measurement)*.* Computing η_M requires evaluating $G(a, K) = I(p_a; W \mid K)$, which depends on the true joint distribution $P(p_a, W, K)$. The entity cannot compute its own η_M; it would need the very information it is trying to learn. This is a structural feature, not a defect: self-knowledge of learning rate would require solving the problem that learning is meant to solve. Both γ and η_M are measurable by an external observer with access to the entity's trajectory and outcomes.

Definition 5 (Intelligence)**.** For an entity E with agency γ and learning efficiency η_M:

$$I(E) = \gamma \cdot \eta_M$$

This is stated here as a definition. Theorem 3 establishes that it is the unique measure consistent with the structural invariants of persistence.

Definition 6 (Agency)**.** Agency $\gamma \in [0, 1]$ is the probability that the entity engages its adaptive capacity M in response to a learning signal. Equivalently, γ is the response gate: the fraction of environmental signals that trigger an update to (A, K).

The operational signature of $\gamma > 0$ is that the entity bounds uncertainty beyond what immediate survival requires, freeing context capacity for novel states. An entity with $\gamma = 0$ responds to no signal and accumulates no free context. An entity with $\gamma > 0$ engages M on signals, resolves uncertainty, promotes actions to lower encoding levels, and thereby maintains $C_{\text{free}} > 0$ (Definition 2).

The distinction between $\gamma = 0$ and $\gamma > 0$ is not between entities that face threats and those that do not. It is between entities that build reserve capacity for states they have not yet encountered and those that do not.

3 Type Irreducibility of A and M

Theorem 1 (Type Irreducibility of A and M)**.** *No element of A can become an element of M through any learning process. No element of M can become an element of A through any learning process. The two spaces are type-irreducible.*

Proof. The type signatures are distinct:

$$A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$$
$$M : (A, K) \times L \to (A, K)$$

A takes knowledge and observations as input and returns an action distribution. M takes an action–knowledge pair and a learning signal as input and returns an updated action–knowledge pair. The domains and codomains are different types.

No learning process changes a function's type signature. An element $a \in A$ that achieves certainty (its output concentrates on a single action for all inputs) promotes to a lower encoding level in the hierarchy established in [16], but this is a change of encoding level, not a change of type. The promoted a still maps $K \times W_{\text{obs}} \to \Delta\mathcal{A}$; it is still an element of A.

Conversely, M takes (A, K) as input. An element of A does not accept (A, K) as input; it accepts $K \times W_{\text{obs}}$. No amount of learning can change what a function accepts as input. The type distinction is categorical, not parametric.

Why unification fails. A natural objection holds that the type distinction is definitional: we defined A and M to have different signatures, so of course they are different. The non-trivial content is that any entity attempting to collapse A and M into a single type T faces an impossibility. Suppose $T : K \times W_{\text{obs}} \times (T, K) \times L \to \Delta\mathcal{A} \times (T, K)$. An element of T must simultaneously accept observations (to select actions) and accept its own space as input (to modify itself). But self-application $T(T)$ is ill-typed in the simply-typed lambda calculus [2]. Polymorphic systems (System F, the Calculus of Constructions) permit forms of self-application via universal quantification, but the quantified function does not modify its own type—it operates uniformly across types. The relevant constraint is not syntactic self-application but *self-modification at the same meta-level*: a function that rewrites its own implementation while executing. This requires stratification—exactly the A/M separation.

An anticipated objection: programs routinely modify themselves (`eval`, metaclasses, Lisp macros). But in every such case, the code being modified operates at a lower meta-level than the code doing the modification. The interpreter that evaluates `eval` is not the code being evaluated. This is the A/M separation implemented in software: the runtime (M) modifies the program (A), and these occupy different levels of the execution hierarchy. Languages that appear to collapse the levels achieve it by implicit stratification, not by genuine self-application. The irreducibility is not our assumption but a consequence of type-theoretic consistency. □

Corollary 2 (Distinct Roles)**.** *The type irreducibility establishes distinct functional roles:*

- *A navigates W given K: selecting actions in the current world state using current knowledge.*
- *M evolves (A, K): modifying the entity's action repertoire and knowledge base in response to learning signals.*

Concrete instances of M acting on different targets:

- ***Neuroplasticity***: *M acting on A: modifying which actions are available and how they respond to inputs.*
- ***Insight***: *M acting on K: restructuring the knowledge base to incorporate a new proposition that changes the graph topology.*
- ***Habit formation***: *M promoting from L_n to lower encoding levels, the process described in [16], Theorem 4.*
- ***Evolution***: *M acting on M itself, the highest level of the hierarchy, operating at the most stringent rigor threshold.*

4 Necessity of γ and η_M

Definition 7 (Persistence)**.** An entity E *persists* if it survives for all t: there is no finite T at which $U(w_T, K(T)) > U^d_{\text{lethal}}$ terminates the entity. Persistence is indefinite survival, not survival to some fixed horizon.

Theorem 3 (Structural Invariants of Persistence)**.** *For any entity E persisting (Definition 7) under bounded active context C_n, survival threshold U^d_{lethal}, with positive novelty rate $\nu(E) > 0$ (Definition 1):*

(i) $\gamma(E) > 0$ necessarily.

(ii) $\eta_M(E) > 0$ necessarily (in expectation over the entity's trajectory).

(iii) Every other scalar property of E except $\sigma > 0$ (sensing) can equal zero for some persistent entity.

(iv) Therefore $I(E) = \gamma \cdot \eta_M$ is the natural measure of intelligence, identified by what survival requires.

When $\nu(E) = 0$, the entity faces no novel states and $\gamma = 0$ is viable: no free context is needed because nothing unexpected arrives. Such entities persist without intelligence. Intelligence is what positive novelty rate selects for.

Scope note. This theorem establishes $\gamma > 0$ and $\eta_M > 0$ as *necessary* conditions for persistence. The proof is a selection argument: entities lacking either property are eliminated. The theorem does *not* explain the *origin* of γ: how $\gamma > 0$ structures bootstrap from $\gamma = 0$ substrates is an open question (see Section 8).

Scope clarification. The theorem applies to entities with the structure $E = (K, A, \sigma, \gamma)$. This is a presupposition, not a conclusion: the theorem does not derive the *existence* of agency from non-agentive precursors. What it establishes is an *elimination result*: among entities that have this structure, those with $\gamma = 0$ do not persist under $\nu > 0$. The contribution is identifying *why* $\gamma = 0$ is untenable—the free-context argument—not explaining how $\gamma > 0$ arises from $\gamma = 0$ substrates (which remains an open question; Section 8). Inert objects (rocks, tables) lack this structure entirely and are outside the theorem's domain.

Proof. **Part 1: Necessity of $\gamma > 0$: The Free Context Argument.**

Suppose $\gamma(E) = 0$. By Definition 6, the entity never engages M: its adaptive capacity is zero. Therefore $(A(t), K(t)) = (A(0), K(0))$ for all t; the entity's action–knowledge pair is fixed at its initial state.

The entity has positive novelty rate $\nu(E) > 0$ (Definition 1): it encounters world states where $U(w, K) > 0$ at a positive rate. Each novel state consumes active context capacity. With $\gamma = 0$, no learning occurs: no uncertainty is resolved, no action is promoted to a lower encoding level, no context is freed. $C_{\text{free}}(t) \to 0$ monotonically as unresolved content accumulates. When $C_{\text{free}} = 0$, the entity cannot load the relevant subgraph $K^{[a]}$ for any new problem a: inference quality $\rho \to 0$. The entity acts without relevant knowledge. It encounters w where $U(w, K(0)) > U^d_{\text{lethal}}$ and does not survive.

The argument is self-contained at the individual level and requires no competition. For any initial $K(0)$, because $\nu > 0$ the entity encounters novel states at positive rate. Each novel state consumes context capacity that is never freed: freeing context requires either promoting actions to lower encoding levels or resolving uncertainty, both of which require engaging M [14, 16]. With $\gamma = 0$, no such engagement occurs. Therefore there exists a finite time T^* at which $C_{\text{free}}(T^*) = 0$. After T^*, the entity cannot process any novel state. Since $\nu > 0$ continues to present states where $U(w, K(0)) > 0$, the entity eventually encounters w with $U(w, K(0)) > U^d_{\text{lethal}}$ and does not survive. No competitor is required: the environment alone is sufficient.

The argument extends to other scales:

- **Lineage**: a $\gamma = 0$ lineage disperses copies with fixed $(A(0), K(0))$ across W. Each copy faces the individual argument above; the lineage contracts as copies fail.
- **Institutional**: organizations with $\gamma = 0$ at the organizational level face the same free-context exhaustion.

The critical distinction is between $\nu = 0$ and $\nu > 0$. An entity in a zero-novelty environment (a deep-sea vent organism whose K covers all reachable states, a thermostat in a stable room) faces no context pressure and persists with $\gamma = 0$. The framework does not predict it dies. It predicts it does not need intelligence. The necessity of $\gamma > 0$ is conditional on the environment presenting states the entity has not yet bounded.

Positive entropy in W is the upstream cause of positive novelty rate: energy gradients produce states not contained in any finite K [5, 1]. But entropy is a property of the environment; novelty rate is a property of the entity–environment coupling. Entropy guarantees that $\nu > 0$ eventually for any entity whose niche is not permanently isolated from the rest of W. The "eventually" matters: a $\gamma = 0$ entity in a temporarily stable niche persists until entropy-driven change reaches it. The theorem's premise $\nu > 0$ is the condition that change has arrived.

Therefore $\gamma(E) > 0$ is necessary for persistence under $\nu > 0$.

Part 2: Necessity of $\eta_M > 0$.

Suppose $\gamma > 0$ but $\eta_M \leq 0$.

Case $\eta_M = 0$: The entity has drive ($\gamma > 0$) but M produces no improvement in grounding, conditional on engagement. (A, K) is effectively fixed despite engagement. As in Part 1, the free-context exhaustion argument applies: novel states consume context that is never freed by promotion, and $\rho \to 0$.

As with γ, the exception is $\nu = 0$: an entity whose $K(0)$ already covers all reachable states faces no novel demands and persists with $\eta_M = 0$. The necessity of $\eta_M > 0$ is conditional on $\nu > 0$, exactly as for γ.

Additionally, C_n is bounded. Novel world states generate learning signals that fill the active context L_n with unresolved content. By the escalation result [14], inference quality ρ degrades as L_n fills without promotion to lower encoding levels. Promotion requires $\eta_M > 0$: achieving the rigor threshold θ_i at encoding level i requires M to improve the relevant action to the required certainty (Theorem 4 of [16]). With $\eta_M = 0$, no action reaches threshold, L_n is never relieved, and $\rho \to 0$. Inference fails.

Case $\eta_M < 0$ persistently: (A, K) worsens in expectation with each learning event. Grounding decreases; uncertainty increases. This accelerates the failure mode of the $\eta_M = 0$ case.

The persistence condition is:

$$\mathbb{E}_t[\eta_M] > 0 \quad \text{over the entity's trajectory}$$

not $\eta_M(t) > 0$ at every time step. Temporary $\eta_M < 0$ (exploration, creative destruction) is compatible with persistence and can increase long-run survival. What is required is that the trajectory-averaged learning efficiency is positive.

Therefore $\mathbb{E}_t[\eta_M] > 0$ is necessary for persistence.

Part 3: Minimality: every other property can equal zero.

For every other scalar property of E, there exists a persistent entity for which that property equals zero:

- $K(0) = \varnothing$: an entity with $\gamma > 0$ and $\eta_M > 0$ survives starting from empty knowledge; it builds K from monotone descent on $H(W \mid K)$ [16].
- $A(0) = \{\text{id}\}$: the identity action still returns feedback from W, grounding the first proposition.
- $C_n = C_{\min}$: minimal context capacity. Graph structure [13] maintains $\rho = 1$ by retrieving exactly the relevant subgraph per problem.
- Substrate, observable window W_{obs}: these are free parameters. They constrain the rate and form of learning but not its possibility.

The exception is sensing: $\sigma > 0$ is necessary because $\gamma > 0$ requires learning signals, and learning signals require sensing. An entity with $\sigma = 0$ receives no observations from W, M receives no input, and no update occurs regardless of γ. However, $\sigma > 0$ is a precondition for the entity to be an entity in the relevant sense: without sensing, there is no W_{obs}, no feedback, and no action–world coupling. The theorem's domain (Definition 6) presupposes $\sigma > 0$ as part of entity structure, not as a separate invariant.

No other scalar property has the survival-critical necessity established in Parts 1 and 2.

Part 4: $I(E) = \gamma \cdot \eta_M$ follows by arithmetic.

Parts 1–3 are the substantive content: γ and η_M are the unique structural invariants of persistence. Given that γ is a probability and η_M a conditional rate, their product is the expected improvement rate by the law of total expectation (Corollary 4). The discovery is in Parts 1–3; the product form is a consequence of what the invariants *are*, not a separate result. □

5 $I(E) = \gamma \cdot \eta_M$ Is Structurally Necessary

Corollary 4 (The Intelligence Measure)**.** *γ and η_M are the unique structural invariants of persistence (Theorem 3). The expected improvement rate of the entity is:*

$$I(E) = \gamma \cdot \eta_M$$

This is not selected from a family of candidates but derived from the types of the invariants: γ is a probability; η_M is a conditional rate. Their product is the expected rate by the law of total expectation.

Proof. $\gamma \in [0, 1]$ is the marginal probability of engaging M. η_M is the expected improvement rate *conditional on engagement* (Definition 4). By the law of total expectation:

$$\mathbb{E}[\text{improvement rate}] = \gamma \cdot \mathbb{E}[\text{rate} \mid \text{engaged}] + (1 - \gamma) \cdot 0 = \gamma \cdot \eta_M$$

This identity holds whether γ is content-blind (a fixed coin flip) or directed (the entity selectively engages on promising signals). The conditioning in Definition 4 absorbs the selection effect: η_M already reflects whichever signals the gate admits. The product is exact, not an approximation.

Many functions $g(\gamma, \eta_M)$ are zero on the axes and positive in the interior ($\gamma^2 \cdot \eta_M$, $\gamma \cdot \eta_M^2$, etc.). What distinguishes the product is not a uniqueness argument over function space but the *types of the operands.* γ is a probability of engagement; η_M is the rate conditional on engagement. Total expectation gives the unconditional rate. Any other combination would require interpreting γ or η_M as something other than what they are. The measure is derived from the structure of the invariants, not stipulated.

The product $\gamma \cdot \eta_M$ has the same structure as the retention criterion of Theorem 1 in [15]: probability of needing it $\times$ conditional value given need. Both are instances of expected value under a binary gate. □

Remark 2 (Exploration and Creative Destruction)*.* Temporarily negative η_M, locally irrational steps that reduce grounding in the short run, is survival-optimal when the persistence condition requires only $\mathbb{E}_t[\eta_M] > 0$ over the trajectory, not $\eta_M \geq 0$ at every step. Exploration (trying actions with uncertain outcomes) and creative destruction (dismantling existing (A, K) structure to rebuild it better) are the signature of M sophisticated enough to optimize over trajectories rather than individual steps. An entity restricted to $\eta_M \geq 0$ at every step is trapped in local optima; an entity that can tolerate temporary $\eta_M < 0$ accesses a larger region of the (A, K) landscape.

6 Engagement with Prior Work

Dupoux, LeCun, and Malik [6]. They deliberately avoid defining intelligence, instead characterizing it as the capacity for autonomous learning. Theorem 3 establishes that this is structurally correct: $\gamma > 0$ (autonomous engagement) and $\eta_M > 0$ (learning efficiency) are the unique invariants. Their product is the definition they avoided giving. Their dual-system architecture (observation + action with meta-control routing) corresponds to the A/M distinction established in Theorem 1, with meta-control corresponding to $M^{(2)}$.

Hutter [11] and Legg–Hutter [12]. They define intelligence as average performance over computable environments, formalized via AIXI. Three structural differences from the present account:

(1) AIXI assumes unbounded computation; we have bounded C_n. (2) They define intelligence as performance level; we define it as rate of improvement (η_M), independent of current performance. A novice chess player learning rapidly has higher η_M than a grandmaster at plateau, despite lower performance. (3) Their environment class is all computable functions; ours is physical W with positive entropy, a narrower but physically grounded class.

Chollet [4]. Skill-acquisition efficiency, the core of the Abstraction and Reasoning Corpus, is structurally close to η_M. Two differences: Chollet treats prior knowledge K as a confound to be controlled for in measurement; here K is constitutive, because η_M operates on (A, K) and the efficiency of learning depends on what is already known. Chollet's "scope" and "generalization difficulty" map to the growth of the reachable territory $T_{\text{reachable}}$ in the (A, K) landscape.

Schmidhuber [19]. Curiosity as compression progress corresponds precisely to η_M: the agent seeks states that maximize improvement in its world model. The difference is foundational. Schmidhuber treats intrinsic drive as a design objective, a reward signal to be engineered into artificial agents. Theorem 3, Part 1, establishes $\gamma > 0$ as a necessary condition of persistence: not a reward to be designed in, but a property that selection enforces.

Friston [9, 10]. The free energy principle derives minimization behavior from self-organization: systems that persist must minimize surprise (free energy). This is consistent with monotone descent on $H(W \mid K)$ [16]. The gap is the minimization *imperative*: FEP establishes that self-organizing systems *do* minimize free energy, but does not establish why $\gamma > 0$ is structurally necessary: why entities that fail to minimize are eliminated. Theorem 3, Part 1, fills this gap via the lineage argument: $\gamma = 0$ lineages contract under positive novelty rate $\nu(E) > 0$ regardless of their initial fitness.

Cattell [3] and Spearman [23]: Psychometric intelligence. Cattell's distinction between fluid intelligence (G_f) and crystallized intelligence (G_c) maps onto the framework. G_f, the capacity for novel problem-solving independent of prior knowledge, is structurally close to η_M: the rate at which the entity extracts grounding from new learning signals. G_c, accumulated knowledge and skill, is $|K|$, the size of the knowledge base. The framework predicts the well-established correlation between G_f and G_c: entities with higher η_M build larger K faster, so the two measures co-vary despite measuring different quantities.

Spearman's g-factor, the dominant factor across psychometric tests , should, on this account, correlate with $I(E) = \gamma \cdot \eta_M$. This is a testable prediction: g should load on both engagement rate (willingness to attempt novel problems) and learning efficiency (improvement per attempt), and the product should predict g better than either factor alone.

7 The Complete Chain

The five papers derive the complete architecture of intelligence from a single generative constraint. Each step is a consequence of bounded active context interacting with properties of the physical world:

1. **Bounded C_n + information preservation $\Rightarrow$ graph structure** [13]. Any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges.

2. **Bounded C_n + growing A $\Rightarrow$ convergence to escalation** [14]. Any viable control architecture converges toward escalation properties as concurrent demands grow relative to C_n.

3. **Survival pressure $\Rightarrow$ K/A inseparability with asymmetric retention** [15]. Under survival pressure, knowledge and action become inseparable; the retention criterion is probability of need $\times$ conditional value.

4. $\gamma > 0$ **+ feedback from** W $\Rightarrow$ **gradient descent on** U **with convergence; cost-of-failure $\Rightarrow$ encoding hierarchy** [16]. Agency with feedback produces convergent uncertainty reduction. Asymmetric costs generate a permanent encoding hierarchy.

5. **Positive entropy + bounded** C_n $\Rightarrow$ $\gamma > 0$ **and** $\eta_M > 0$ **necessary;** $I(E) = \gamma \cdot \eta_M$ **unique measure** [this paper]. The lineage argument and the context-filling argument establish the two invariants. Their product is the unique intelligence measure.

Each step is a consequence of the same generative constraint. The contribution is not any individual result but the connected chain: from bounded context to graph structure to survival-driven retention to escalation to evaluation-driven descent to the structural definition of intelligence.

8 Open Questions

1. **Origin of γ.** Theorem 3 establishes that $\gamma > 0$ is necessary for persistence but does not explain how $\gamma > 0$ structures bootstrap from substrates where $\gamma = 0$. The thermodynamic conjecture: dissipative structures [18, 7] under sustained energy flow spontaneously produce $\gamma > 0$ configurations, which selection then retains. This is distinct from the necessity result.

2. **Direction of γ.** Is agency uniformly distributed across learning signals, or does it have intrinsic directedness? If directed, what determines the direction?

3. **The $M^{(n)}$ hierarchy.** How many levels of meta-learning does a persistent entity require? Is there a natural termination point, or is the hierarchy open upward without bound?

4. **Bound composition in A.** How do individual action bounds compose when multiple actions share knowledge substructure?

9 Conclusion

Intelligence is the driven capacity to improve. This is not proposed but identified by elimination. The lineage argument (Theorem 3, Part 1) establishes that $\gamma = 0$ is untenable: entities without agency contract under positive novelty rate $\nu(E) > 0$ and are eliminated by selection. The context-filling argument (Part 2) establishes that learning efficiency $\eta_M > 0$ is necessary: without improvement, bounded context fills with unresolved content and inference degrades to failure. Part 3 establishes minimality: every other property of a persisting entity can equal zero for some persistent entity. The product $I(E) = \gamma \cdot \eta_M$ is the unique measure consistent with these invariants, derived from the law of total expectation applied to a probability gate and a conditional rate.

The type irreducibility of A and M (Theorem 1) ensures that the distinction between executing and learning is not an implementation choice but a categorical necessity: no learning process can transform an action into a learning process or vice versa.

The five-paper chain derives, from the single constraint of bounded active context, the complete architecture: graph structure, knowledge–action inseparability, escalation dynamics, evaluation-driven descent with encoding permanence, and the structural invariants of intelligence. Each result

is a necessary consequence of the generative constraint interacting with properties of the physical world: positive entropy, survival pressure, and feedback. Nothing is stipulated; everything follows.

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References

[1] W. R. Ashby. *An Introduction to Cybernetics*. Chapman and Hall, 1956.

[2] H. P. Barendregt. *The Lambda Calculus: Its Syntax and Semantics*. North-Holland, revised edition, 1984.

[3] R. B. Cattell. *Abilities: Their Structure, Growth, and Action*. Houghton Mifflin, 1971.

[4] F. Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

[5] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.

[6] E. Dupoux, Y. LeCun, and J. Malik. Why AI systems don't learn and what to do about it: Lessons on autonomous learning from cognitive science. *arXiv preprint arXiv:2603.15381*, 2026.

[7] J. L. England. Statistical physics of self-replication. *The Journal of Chemical Physics*, 139(12):121923, 2013.

[8] C. Finn, P. Abbeel, and S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1126–1135, 2017.

[9] K. Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[10] K. Friston. Active inference and learning. *Neuroscience and Biobehavioral Reviews*, 68:862–879, 2017.

[11] M. Hutter. *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, 2005.

[12] S. Legg and M. Hutter. Universal intelligence: A definition of machine intelligence. *Minds and Machines*, 17(4):391–444, 2007.

[13] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Manuscript, 2026.

[14] P. D. McCarthy. Convergence of control architectures to escalation under bounded context. Manuscript, 2026.

[15] P. D. McCarthy. Knowledge and action are inseparable under survival pressure. Manuscript, 2026.

[16] P. D. McCarthy. Evaluation-driven state revision and the encoding hierarchy under bounded context. Manuscript, 2026.

[17] J. Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, 1988.

[18] I. Prigogine and I. Stengers. *Order Out of Chaos: Man's New Dialogue with Nature*. Bantam Books, 1984.

[19] J. Schmidhuber. Formal theory of creativity, fun, and intrinsic motivation (1990–2010). *IEEE Transactions on Autonomous Mental Development*, 2(3):230–247, 2010.

[20] E. Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.

[21] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[22] H. A. Simon. A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.

[23] C. Spearman. "General intelligence," objectively determined and measured. *The American Journal of Psychology*, 15(2):201–292, 1904.

[24] S. Thrun and T. M. Mitchell. Learning to learn. In *Learning to Learn*, pages 3–17. Springer, 1998.